

UK Maths Trust

SENIOR MATHEMATICAL CHALLENGE

Thursday 9 October 2025

Organised by the United Kingdom Mathematics Trust

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MARKETS

*Candidates must be full-time students at secondary school or FE college.
England & Wales: Year 13 or below | Scotland: S6 or below | Northern Ireland: Year 14 or below*

INSTRUCTIONS

1. Do not open the paper until the invigilator tells you to do so.
2. Time allowed: **90 minutes**. No answers, or personal details, may be entered after the allowed time is over.
3. The use of blank paper for rough working is allowed; **squared paper, calculators and measuring instruments are forbidden**.
4. **Use a B or an HB non-propelling pencil**. Mark at most one of the options, A, B, C, D, or E, on the Answer Sheet for each question. Do not mark more than one option.
5. **Do not expect to finish the whole paper in the time allowed**. The questions in this paper have been arranged in approximate order of difficulty with the harder questions towards the end. You are not expected to complete all the questions during the time. You should bear this in mind when deciding which questions to tackle.
6. **Scoring rules**: All candidates start with 25 marks; 0 marks are awarded for each question left unanswered; 4 marks are awarded for each correct answer; 1 mark is deducted for each incorrect answer (to discourage guessing).
7. **Your Answer Sheet will be read by a machine**. Do not write or doodle on the sheet except to mark your chosen options. The machine will read all black pencil markings even if they are in the wrong places. If you mark the sheet in the wrong place, doodle, or leave bits of eraser stuck to the page, the machine will interpret the mark in its own way, or reject the answer sheet.
8. **The questions on this paper are designed to challenge you to think, not to guess**. You will gain more marks, and more satisfaction, by doing one question carefully than by guessing lots of answers. This paper is about solving interesting problems, not about lucky guessing.
9. To accommodate candidates sitting at other times, please do not discuss the paper on the internet until 08:00 BST on Saturday 11th October, when the solutions video will be released at ukmt.org.uk/video-solutions-list. Candidates in time zones more than 5 hours ahead of GMT must sit the paper on Friday 10th October (as defined locally).

Enquiries about the Senior Mathematical Challenge should be sent to:

challenges@ukmt.org.uk www.ukmt.org.uk

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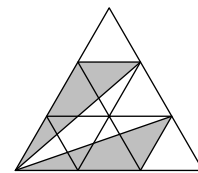
1. Pablo has 100 identical small cubes. He uses some of them to build the largest possible solid cube. How many of the small cubes are left over?

A 16 B 27 C 36 D 73 E 92

2. The diagram shows an equilateral triangle divided into nine smaller equilateral triangles, with two additional lines.

What fraction of the large triangle is shaded?

A $\frac{1}{3}$ B $\frac{1}{4}$ C $\frac{3}{8}$ D $\frac{2}{9}$ E $\frac{4}{9}$



3. What is $25^2 - 24^2 - 23^2 + 22^2$?

A 0 B 1 C 2 D 3 E 4

4. Sonia writes down three 2-digit numbers whose sum is 46. The first number is prime, the second is square and the third is even.

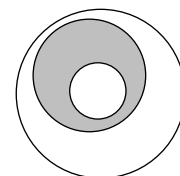
What is the even number?

A 10 B 12 C 14 D 16 E 18

5. The diagram shows three circles with radii 1, 2 and 3.

What is the ratio of the shaded area to the area of the largest circle?

A 1 : 3 B 1 : 2 C $\sqrt{2} : \sqrt{3}$ D 2 : 3 E 4 : 9



6. For what value of x is $\sqrt{(\sqrt{(\sqrt{x} + 1)} + 1) + 1} = 3$?

A 4096 B 64 C 8 D 3 E 0

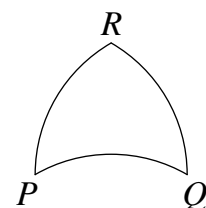
7. What is the 100th term of the sequence 1, 5, 7, 11, 13, 17, 19, 23, ... whose terms are consecutive odd numbers but with all the multiples of 3 removed?

A 201 B 203 C 247 D 299 E 301

8. The diagram shows three arcs of circles of radius 1. P is the centre of the circle of which RQ is an arc and Q is the centre of the circle of which PR is an arc.

What is the area of this shape?

A $\frac{\sqrt{3}\pi}{10}$ B $\frac{\pi}{6}$ C $\frac{\pi}{8}$ D $\frac{\sqrt{3}}{4}$ E $\frac{\sqrt{2}\pi}{9}$



9. Fifty squares are drawn side by side in a line. The first and last squares are shaded. Other squares in the line must be shaded such that both these rules apply: (a) no two adjacent squares are shaded and (b) there are no more than three consecutive unshaded squares.

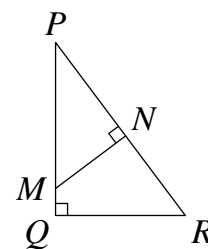
What is the difference between the smallest and largest number of squares that can be shaded?

A 8 B 10 C 11 D 13 E 15

10. How many different squares are factors of 2025?

A 2 B 3 C 4 D 5 E 6

11. The diagram shows a triangle PQR with $\angle PQR = 90^\circ$, $PQ = 20$ and $PR = 25$. Point M lies on PQ , point N lies on PR and PNM is a right-angled triangle whose area is half that of triangle PQR .



What is the length of MN ?

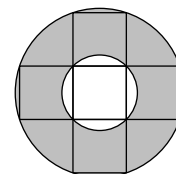
- A $6\sqrt{2}$ B $\frac{15}{2}\sqrt{2}$ C $8\sqrt{2}$ D $\frac{17}{2}\sqrt{2}$ E $10\sqrt{2}$

12. Ayain writes down all the 3-digit numbers consisting of three different odd digits.

How many of Ayain's numbers are divisible by 3?

- A 42 B 36 C 30 D 24 E 18

13. Five congruent squares, each of side $2a$, are placed edge to edge. Two circles with the same centre are drawn through the vertices as shown.



What is the area of the region between the two circles?

- A $2\pi a^2$ B $4\pi a^2$ C $6\pi a^2$ D $8\pi a^2$ E $10\pi a^2$

14. The simultaneous equations $x + \frac{1}{y} = 2$ and $y + \frac{1}{x} = \frac{9}{4}$ have two pairs of real solutions.

What is the difference between the possible values of x ?

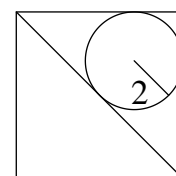
- A $\frac{1}{3}$ B $\frac{2}{3}$ C $\frac{3}{4}$ D 1 E $\frac{3}{2}$

15. The integer n is such that $1 < n < 99$. P is the difference between 1 and n , Q is the difference between 99 and n , and R is the difference between P and Q .

For how many values of n is R a prime number?

- A 0 B 2 C 4 D 8 E 99

16. The diagram shows a square, one of its diagonals and a circle. The circle touches the diagonal and two sides of the square. The circle has radius 2.

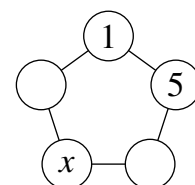


What is the length of the side of the square?

- A $4 + 2\sqrt{2}$ B $8 - \sqrt{3}$ C $2 + 2\sqrt{2}$ D $8 - 2\sqrt{2}$ E $4 + 2\sqrt{3}$

17. The circles are filled with five integers so that any integer from 1 to 21 can be made, either by choosing one of the integers or by summing up to 5 adjacent integers.

When 1 and 5 are in the positions shown, what is the value of x ?



- A 2 B 3 C 7 D 10 E 11

18. The positive integer N has 2025 digits. The first digit is a 3. Every two consecutive digits of N form a number that is divisible by either 17 or 23. The units digit of N could either be p or q .

What is the value of $p + q$?

- A 3 B 6 C 7 D 9 E 10

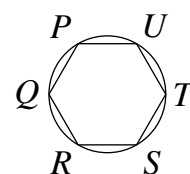
19. A 'complete' football kit consists of a shirt, a pair of shorts and a pair of socks. Three pairs of shorts and one pair of socks together cost the same as two shirts. Seven pairs of shorts and four pairs of socks together cost the same as five shirts. Eden has exactly the right amount of money to buy nine shirts. How many 'complete' football kits could be bought for the same amount of money?

A 3 B 4 C 5 D 6 E 7

20. When $\frac{1}{x} - \frac{1}{y} = 2025$, what is the value of $\frac{x + 2026xy - y}{2y - 2025xy - 2x}$?

A 0 B 1 C $\frac{1}{2}$ D 2026 E $\frac{1}{2025}$

21. A regular hexagon $PQRSTU$ is inscribed in a circle of radius 5. A point X on the circumference of the circle is connected to the vertices of the hexagon to form six chords XP , XQ , XR , XS , XT and XU .



What is the value of $XP^2 + XQ^2 + XR^2 + XS^2 + XT^2 + XU^2$?

A 150 B 216 C 256 D 300 E 360

22. Together, n aardvarks and 12 anteaters eat $n^2 + 20n + 25$ ants. Each animal eats the same whole number of ants.

How many ants does each animal eat?

A 61 B 66 C 71 D 74 E 79

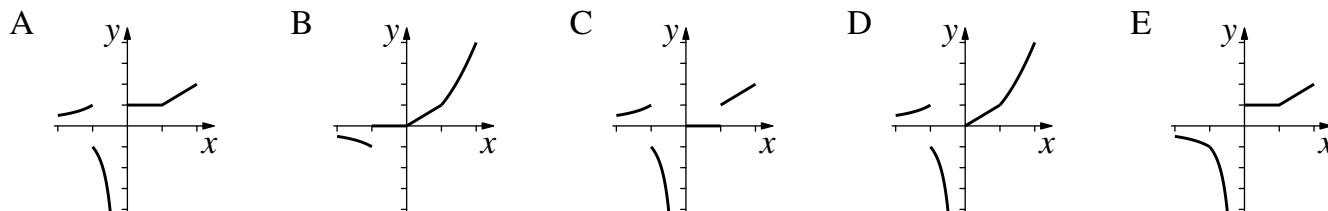
23. Jemima took a series of n tests each with the same maximum mark. After $(n - 2)$ tests her average score was m . She scored full marks in the $(n - 1)$ th test, and so raised her average score by 4 marks. In the n th test when she again scored full marks, she increased her average score by 3 marks.

How many tests did Jemima take?

A 6 B 7 C 8 D 9 E 10

24. The output of the function $F(x)$ when applied to a real number x is the greatest integer less than or equal to x . For example, $F(3) = 3$, $F(4.7) = 4$, $F(-2.3) = -3$.

Which of the following is the graph of $y = x^{F(x)}$ for non-zero values of x in the interval $-2 < x < 2$?



25. The diagram shows two squares, $OPQR$ and $OSTU$. Point S lies on QV . Triangle UQV is isosceles with a right angle at V . Square $OPQR$ has area 25.

What is the area of square $OSTU$?

A 36 B 45 C 50 D 54 E 60

